CardioTwin — System Architecture

Personalized Cardiovascular Digital Twin for Proactive Healthcare
Synthetic-data research prototype; not clinically validated.

End-to-end flow

Synthetic EHR generator + **Wearable simulator** + **Explicit event module**
→ **Leakage-safe feature engineering**
→ **Patient-level ML pipeline**
→ **Validation-only probability calibration + alert threshold**
→ **Saved calibrated model**
→ **FastAPI**
→ **React / Recharts dashboard**

In parallel, prerecorded observations feed a **deterministic replay clock → digital twin state manager → isolated what-if scenario engine**. Prediction and scenario records are written to **SQLite audit storage**.

Component responsibilities

Component	Responsibility
Synthetic EHR	Patient demographics, histories, and generated reference baselines
Wearable simulator	Five-minute HR, HRV, simulated BP, steps, activity, sleep, sensor quality, missingness
Event module	Rule-derived event onset/end timestamps and 60-minute labels; event logs excluded from model features
Feature engineering	Static, trailing-window dynamic, and personal-baseline features using past/current data only
ML	Logistic baseline plus four gradient-boosting ablations; patient-level 70/15/15 split
Calibration	Platt scaling and F1 threshold selection on validation patients only
Digital twin	Profile, current physiology, baseline, rolling features, forecast, quality, and history
Scenario engine	Copies current state and reruns the same model on hypothetical inputs without mutating the original twin
FastAPI	Validated patient, observation, forecast, replay, scenario, history, and evaluation endpoints
SQLite	Local demo persistence for prediction and scenario audit records
React dashboard	Patient selection, trends, calibrated forecast, SHAP explanations, scenarios, ablations, warnings

Methodological guardrails

1. **Patient-level split:** no patient appears in more than one train, validation, or test group.
2. **Past/current-only features:** rolling windows use only information available at the prediction timestamp; no backward fill.
3. **Event isolation:** event logs, event-active flags, hidden simulator flags, and future labels are never model features.
4. **Validation-only calibration:** probability calibration and the prototype alert threshold use validation patients, not test patients.
5. **Scenario isolation:** what-if execution uses a copied state and is explicitly hypothetical, not a causal treatment-effect estimate.
6. **Responsible framing:** all data are synthetic; outputs are not diagnosis, medical triage, treatment advice, or evidence of clinical validity.

Simulator event definition

- **Reference period:** first 12 simulated hours; median valid HR and systolic BP.
- **Heart rate condition:** at least personal reference +22 bpm.
- **Systolic BP condition:** at least personal reference +17 mmHg.
- **Duration:** at least 20 minutes / four consecutive five-minute observations.
- **Forecast label:** 1 for non-event observations during the 60 minutes before event onset.

This target is a simulator-defined physiological pattern, not a clinically validated endpoint.